What Is an Option?

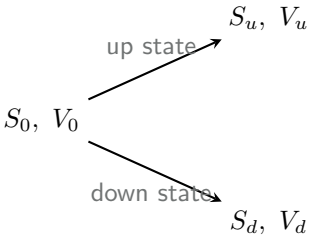
An option gives its holder the right, not the obligation, to transact in an underlying asset on specified terms.

- **Call:** right to buy at the strike price K.
- **Put:** right to sell at the strike price K.
- **Expiration T:** last date the right can be exercised.
- **Premium:** price paid today for the contract.

Central question

How much should someone pay today for a claim that provides a potentially uncertain payoff in the future?

An Arbitrary Financial Claim



Suppose a stock trades today at S_0. At time T, it can be S_u or S_d. A claim V written on the stock then pays V_u or V_d.

Replicating the Financial Claim

Rather than price the claim directly, build a portfolio with exactly the same payoff in every state.

Hold Δ shares of stock and borrow y dollars today at continuously compounded rate r.
Δ is the shares you have bought; y is the amount of the loan.

$$V_u = \Delta S_u - ye^{rT},$$
$$V_d = \Delta S_d - ye^{rT}.$$

Solving for Δ and the Loan

$$V_u - V_d = \Delta(S_u - S_d)$$

$$\therefore \qquad \Delta = \frac{V_u - V_d}{S_u - S_d}$$

$$y = e^{-rT}(\Delta S_d - V_d)$$

Δ is the shares you have bought. y is the amount of the loan.

No-Arbitrage: The Law of One Price

Portfolio lemma

If two portfolios have identical future payoffs in every state, then an arbitrage-free market must assign them the same price today.

$$V_0 = \Delta S_0 - y$$

A One-Step Call Option Example

$S_0 = \$100$, $S_u = \$120$, $S_d = \$80$, $K = \$100$, and $r = 0$. Therefore,

$$V_u = \max(120 - 100, 0) = \$20, \qquad V_d = \max(80 - 100, 0) = \$0,$$

$$\Delta = \frac{20 - 0}{120 - 80} = 0.5, \qquad y = 0.5(80) - 0 = \$40,$$

$$V_0 = 0.5(100) - 40 = \boxed{\$10}.$$

Up state

Half a share is worth $\$60$. Repay $\$40$.
Remaining payoff: $\$20$.

Down state

Half a share is worth $\$40$. Repay $\$40$.
Remaining payoff: $\$0$.

The $\$10$ premium is paid today; $\$20$ or $\$0$ is paid at expiration.

From Replication to Risk-Neutral Pricing

Insert the replicating portfolio into $V_0 = \Delta S_0 - y$:

$$\begin{aligned} V_0 &= \Delta S_0 - e^{-rT}(\Delta S_d - V_d) \\ &= e^{-rT}\left[qV_u + (1-q)V_d\right], \end{aligned}$$

where

$$q = \frac{S_0 e^{rT} - S_d}{S_u - S_d}.$$

Interpretation:

What Are Risk-Neutral Probabilities?

$$S_0 = e^{-rT}\left[qS_u + (1-q)S_d\right]$$

The weights q and $1-q$ make the stock's expected return equal to the risk-free rate.

Actual probabilities

Describe beliefs about what will happen in the world.

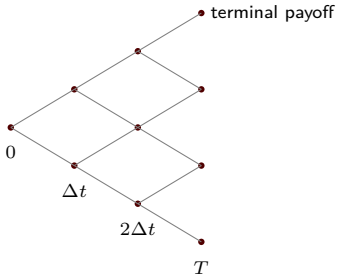
Risk-neutral probabilities

Are pricing weights consistent with traded-asset prices and no-arbitrage.

No-arbitrage condition

$S_d < S_0 e^{rT} < S_u$ ensures $0 < q < 1$. These weights arise from replication, not from forecasting the next stock move.

Extending to Multiple Time Steps



At every terminal node, the contract tells us the payoff. Then work backward:

$$V_t = e^{-r\Delta t}\left[qV^u_{t+\Delta t} + (1-q)V^d_{t+\Delta t}\right].$$

1. Calculate terminal payoffs.
2. Weight and discount adjacent values.
3. Repeat until time zero.

Terminal nodes are model-generated scenarios, not forecasts of realized prices.

From Binomial Trees to Black–Scholes

Discrete binomial model

Stock prices move up or down at individual time steps.

$\rightarrow$

Continuous-time model

Stock prices evolve continuously under a stochastic process.

More steps, smaller Δt

Under an appropriate parameterization, the binomial price converges to the Black–Scholes–Merton price as the number of steps grows.

Black–Scholes–Merton: Equation and Price

Continuous-time delta hedging and no-arbitrage lead to

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

For a European call with no dividends,

$$C = S_0 N(d_1) - K e^{-rT} N(d_2),$$

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

What Does Black–Scholes Actually Tell Us?

It calculates

A theoretical arbitrage-free option premium under its assumptions.

It does not calculate

A prediction of the realized stock price at expiration.

$$C_T = \max(S_T - K, 0),$$
$$P_T = \max(K - S_T, 0).$$

Terminal payoff depends on the unknown S_T. A traded market premium can differ from the model price.

Keep the terms separate

The premium is the upfront price of the contract, not a fee added to its theoretical value.

The Greeks: Sensitivities of Option Value

Delta: $\frac{\partial V}{\partial S}$

Sensitivity to stock price; locally, the shares needed to hedge the option.

Gamma: $\frac{\partial^2 V}{\partial S^2}$

Sensitivity of Delta to the stock price.

Theta

Sensitivity to time; its sign depends on whether time is calendar time or time to expiration.

Vega: $\frac{\partial V}{\partial \sigma}$

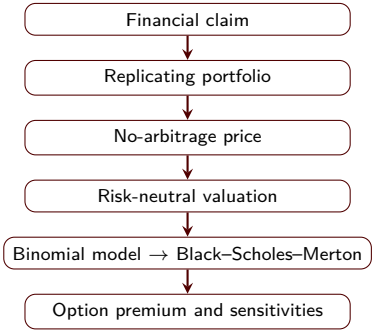
Sensitivity to volatility.

Rho: $\frac{\partial V}{\partial r}$

Sensitivity to the interest rate.

Connection: the Δ from the one-step replicating portfolio becomes a local, continuously adjusted hedge ratio.

Final Takeaway



Central message

Options pricing is not fundamentally about predicting where a stock will go. It is about determining what a financial claim should cost today, given its possible future payoffs and the prices of tradable assets.

Practical Applications: The Options Chain Speaks

Adds missing repos and refreshes history, skipping Options updater, stocks update, and Options Model.

Refresh configured active repos

Run checkup for all teams

Repositories

Add repositories, open one to manage its team, refresh commit history, and inspect cached activity.

Grades cached GitHub activity from the last 7 days for every listed repository team.

org

repo

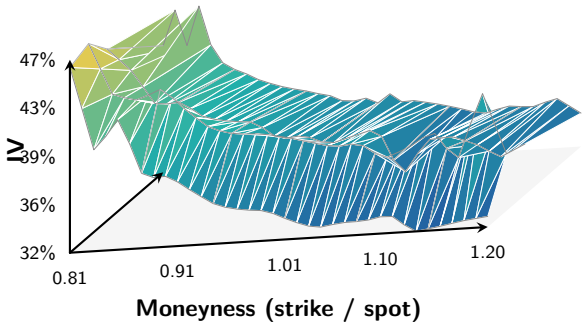
Add repo

AggieInvestmentClub/AIC-BSM active	0 members	0 commits	RED 0/100	No cached commits	Open checkup Markdown PDF
AggieInvestmentClub/Factor-Tree-R-D active	2 members	2 commits	RED 19/100	Latest 2026-10-13 22:12:20	Open checkup Markdown PDF
AggieInvestmentClub/quant-fund-options-ml active	0 members	0 commits	RED 0/100	No cached commits	Open checkup Markdown PDF
AggieInvestmentClub/sellingputsQuantFund active	3 members	5 commits	RED 28/100	Latest 2026-10-13 17:05:52	Open checkup Markdown PDF

Practical Applications: Reading an Options Surface

NVDA implied volatility surface

Oct. 14, 2026 | x: moneyness | depth: expiration (DTE) | height/color: IV



Filtered to 0.80–1.20 moneyness and IV below 150%; interpolated.

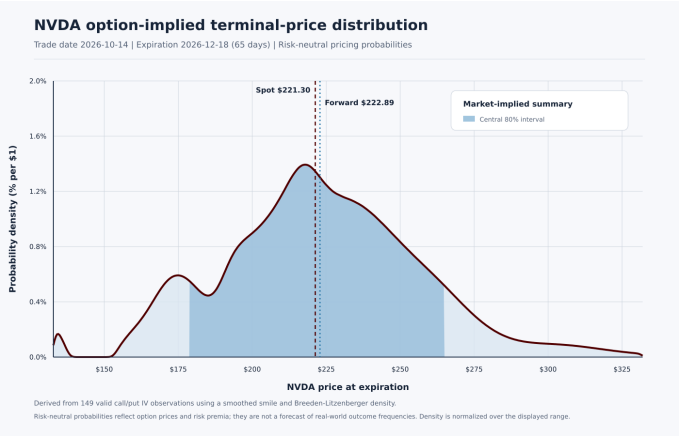
Downside skew

At 0.90 × spot, IV is 3.6–4.5 points above the 1.10 × level. Lower strikes therefore carry higher implied volatility: a downside skew.

Across expirations

At-the-money IV: 33.8% (Nov. 13, 30d), 38.4% (Nov. 20, 37d), 37.2% (Nov. 27, 44d), and 36.6% (Dec. 18, 65d).

From the Volatility Surface to a Market-Implied PDF



How the PDF was generated

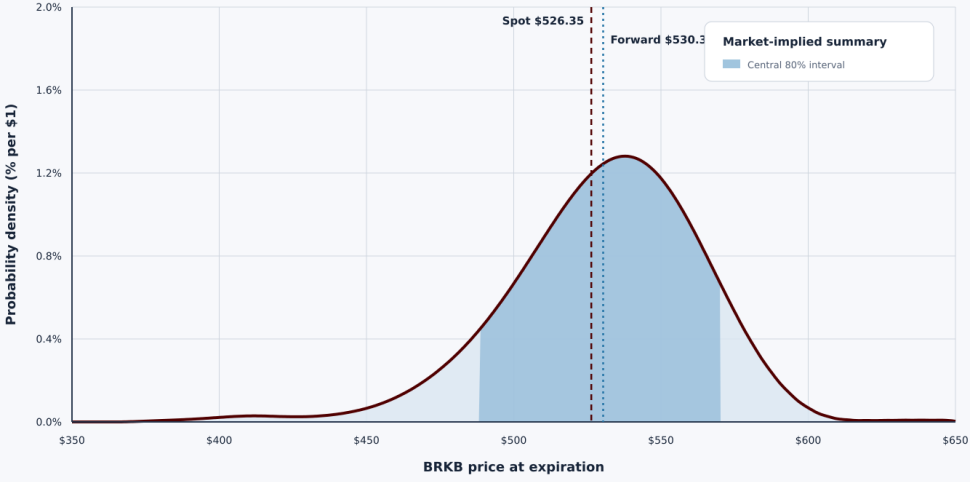
First, smooth implied volatility across strikes and use Black–Scholes to produce a call-price curve $C(K)$.

$$q(K) = e^{rT} \frac{\partial^2 C}{\partial K^2}.$$

Take the second derivative of the call-premium chain across strikes. After adjusting for discounting and normalizing the result, the curvature can be expressed as market-implied probabilities for different NVDA prices at expiration.

BRKB option-implied terminal-price distribution

Trade date 2026-10-14 | Expiration 2026-12-18 (65 days) | Risk-neutral pricing probabilities

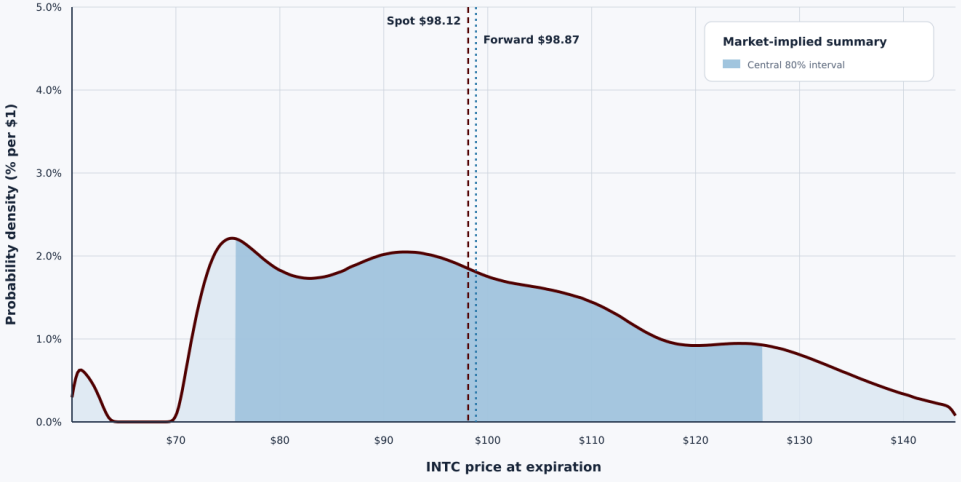


Derived from 52 valid call/put IV observations using a smoothed smile and Breeden-Litzenberger density.

Risk-neutral probabilities reflect option prices and risk premia; they are not a forecast of real-world outcome frequencies. Density is normalized over the displayed range.

INTC option-implied terminal-price distribution

Trade date 2026-10-14 | Expiration 2026-12-18 (65 days) | Risk-neutral pricing probabilities

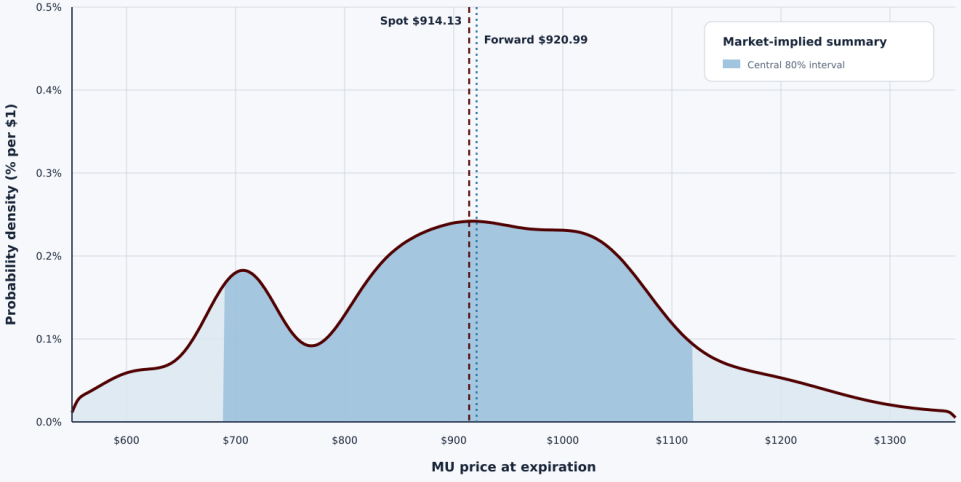


Derived from 51 valid call/put IV observations using a smoothed smile and Breeden-Litzenberger density.

Risk-neutral probabilities reflect option prices and risk premia; they are not a forecast of real-world outcome frequencies. Density is normalized over the displayed range.

MU option-implied terminal-price distribution

Trade date 2026-10-14 | Expiration 2026-12-18 (65 days) | Risk-neutral pricing probabilities

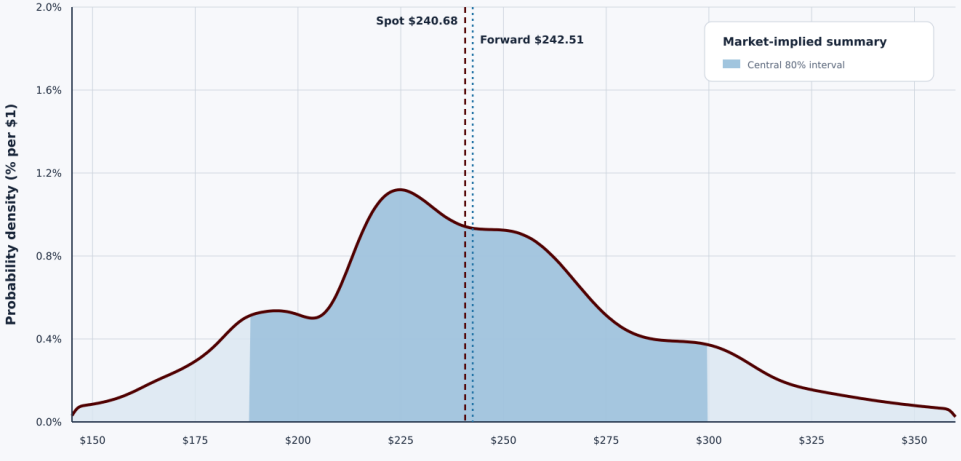


Derived from 122 valid call/put IV observations using a smoothed smile and Breeden-Litzenberger density.

Risk-neutral probabilities reflect option prices and risk premia; they are not a forecast of real-world outcome frequencies. Density is normalized over the displayed range.

ADBE option-implied terminal-price distribution

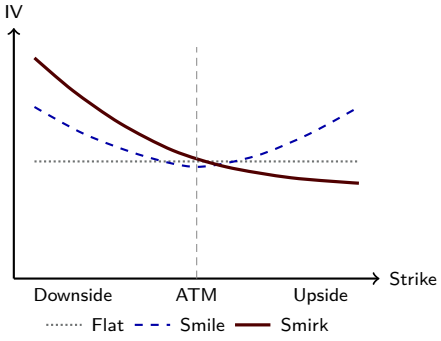
Trade date 2026-10-14 | Expiration 2026-12-18 (65 days) | Risk-neutral pricing probabilities



Derived from 48 valid call/put IV observations using a smoothed smile and Breeden-Litzenberger density.

Risk-neutral probabilities reflect option prices and risk premia; they are not a forecast of real-world outcome frequencies. Density is normalized over the displayed range.

The Volatility Smirk: Why Downside Skew Exists



Same expiration: implied volatility across strikes.

An asymmetric curve

Flat skew assigns similar IV across strikes. A smile raises IV in both tails. A **smirk** is steeper on one side of the distribution than the other.

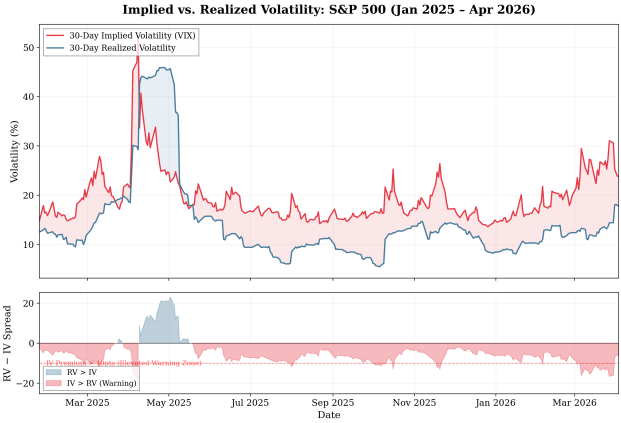
What downside smirk conveys

Market prices assign a larger effect on implied volatility to a downside move than to a comparable upside move. Lower-strike options therefore trade at higher IV.

How does skew exist? Supply meets demand

Investors buy puts to protect portfolios against falling markets, which have historically produced larger and more severe moves. That demand raises downside-option prices and their implied volatilities. At a fundamental level, skew emerges as buyers and sellers meet at different prices across strikes; implied volatility is simply those option prices expressed in volatility terms.

Why Implied Volatility Often Exceeds Realized Volatility



S&P 500, Jan. 2025–Apr. 2026. A negative IV spread means implied volatility exceeded trailing realized volatility.

Demand for options

Traders pay a premium for efficient, leveraged exposure to the underlying asset and, more importantly, protection against adverse price movements.

Compensation for sellers

Option sellers are the counterparties to that demand. They require compensation for bearing volatility risk and the possibility of extreme tail losses.

The volatility risk premium

Together, option demand and seller compensation contribute to implied volatility often exceeding subsequent realized volatility. Academics call this difference the **volatility risk premium**.

Iron Condor: Strike Math and Payoff



SPY example shown: $771.77 spot, 22 DTE, 12.9% implied volatility.

The short-volatility trade

Investors attempt to capture the volatility risk premium by selling options and accepting exposure to volatility and potentially severe market movements. The strategy can generate consistent premiums during relatively stable markets, but may experience substantial losses during periods of extreme volatility.

Four legs, one credit

Buy 762P / sell 767P; sell 777C / buy 782C. Both wings are $5 wide. Net credit and max profit: $3.78 per share, or **$378 per 1-lot**. Max loss: $122.

At expiration

Max profit occurs when SPY finishes between the two short strikes ($767–$777). Beyond either long wing, the loss is capped at $122 per 1-lot.